

ABOUT US:

Project title: Beacon Night

Team name: Four bytes

Hackathon name: Fortex36

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<https://github.com/charanmarpu489-design/beacon-night/tree/main>

Abstract:

- Finding suitable restaurants and lodges is often challenging for families and travelers, especially during night hours and early mornings. Many existing platforms fail to provide accurate information regarding availability, budget suitability, crowd conditions, and family-friendliness.
- This project proposes a web-based solution that helps users locate family-friendly restaurants and lodges based on budget, open/closed status, and busy/not busy conditions.
- The system is designed to prioritize night-time usability, where inconvenience and safety concerns are higher, while still being functional throughout the day.
- By using location-based services and time-based filtering, the platform ensures a reliable and user-friendly experience. This solution reduces time wastage, improves comfort for families, and enhances decision-making during travel.

Problem statement:

- Families and travelers frequently face difficulties in identifying suitable restaurants and lodges that meet their requirements.
- The problem becomes more severe during night-time, when limited options are available and incorrect information leads to frustration and safety concerns.
- Existing applications often lack filters for family-friendly places, budget selection, and crowd status, and they may show places that are closed or overcrowded.
- There is a strong need for a system that provides real-time availability, budget clarity, and crowd awareness to help users make quick and informed decisions, especially during critical hours.

Proposed Solution:

The proposed solution is a web-based platform that allows users to find family restaurants and lodges based on their preferences and real-time conditions.

Users can filter results according to:

- Budget range
- Family-friendly restaurants
- Open or closed status
- Busy or not busy condition

The system uses the current time and location to show only relevant and available options. The platform is optimized for night-time usage, ensuring quick access, clarity, and reliability, while also being useful during mornings and daytime.

Technology stack:

- Frontend: HTML, CSS, JavaScript
 - Used to create a responsive and user-friendly interface
- Backend: Node.js / Firebase (if applicable)
 - Handles data processing and availability logic
- APIs: Location and Map APIs
 - Used for real-time location-based search
- Database: Restaurant and lodge data
 - Stores details such as budget range, family type, and availability

Each technology was chosen for its simplicity, scalability, and effectiveness in delivering real-time results.

System Architecture and design:

- The system follows a client-server architecture.
The user interacts with the frontend through a web browser.
- Location and preference inputs are sent to the backend, where time-based and condition-based filtering is performed.
- The backend fetches relevant restaurant and lodge data and sends the filtered results back to the frontend.
- The architecture ensures fast responses, accurate results, and smooth user experience, especially during urgent night-time scenarios.

Implementation Details:

The frontend was developed using standard web technologies to ensure responsiveness across devices.

Core functionalities include:

- Budget selection module
- Family-restaurant filter
- Open/close status checker based on current time
- Busy/not busy indicator

The backend processes user inputs and applies logic to determine availability and crowd condition before displaying results. The project code is modular, making it easy to extend in the future.

Results and Demonstration:

The developed website successfully displays:

- Nearby family-friendly restaurants
- Budget-based options
- Open and closed status
- Busy and not busy indicators
- Nearby lodges

The system performs efficiently and provides accurate, useful results. The solution demonstrates practical usability, especially during night hours when users need quick and reliable information.

Challenges Faced:

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- Nearby family-friendly restaurants
- Budget-based options
- Open and closed status
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- Nearby lodges

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Future scope:

- Live updates from restaurant and lodge owners
- Table and room booking integration
- User reviews and ratings
- Mobile application development
- AI-based crowd prediction
- Emergency and 24/7 service filters

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Conclusion:

- This project addresses a real-world problem faced by families and travelers by providing a reliable platform to find budget-friendly, family restaurants and lodges based on real-time availability.
- The solution is especially impactful during night-time, reducing inconvenience, improving safety, and saving time.
- The project demonstrates practical implementation, customer-focused design, and strong potential for real-world deployment.